

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
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Performance Guarantee and Test Procedure

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R01	02-06-22	Issued for Client's Review
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A01		

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED- GEN-PRC-001

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEM as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

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Definition of Terms

Terms	Definitions
BFPCL	Brass Fertilizer & Petrochemical Limited.
Employer	BFPCL
COMPANY	BFPCL
GGPL	Gas Gathering Pipelines and Associated Infrastructure
GPP	Gas Processing Plant (at Akabel)
GPMC	Gas Processing and Methanol Complex (at Akabel)
Delivery Point	Custody Transfer Point between SPDC (Gas Seller) and BFPCL (Gas Buyer)
SPDC	Shell Petroleum Development Company Nig. Limited
Contractor	Party employed by COMPANY to carry out the work.
Supplier	Party, which manufactures or supplies equipment and services to perform the duties specified by the Contractor.
OEM	Original Equipment Manufacturer - Party which manufactures equipment to perform the duties specified by the Contractor
Shall	The word 'shall' indicate a mandatory requirement.
Should	The word 'should' indicate a recommendation.
Measuring Range	Range defined by LRL and URL within which the limits of uncertainty of the measuring instrument are specified (the capability of the instrument).
On-Line Instruments	Instruments connected to process and utility lines or equipment via primary isolation valves.
Project	Front End Engineering Design for Gas Gathering Pipelines (Phase I)
NUPRC	National Upstream Petroleum Regulatory Commission

Abbreviations

Abbreviation	Meaning
AG	Associated Gas
ANSI	American National Standards Institute
API	American Petroleum Institute
BBOU	Bubouwe Bou Manifold
BFPCL	Brass Fertilizer & Petrochemical Company Limited
BLV	Blowdown Valve

CAO	Computer Aided Operations
ESD	Emergency Shut Down
FEED	Front End Engineering Design
FLB	Field Logistics Base
FLM	First Line Maintenance
FQI	Flow Quantity Indicator
GSPA	Gas Sales & Purchase Agreement
HP	High Pressure
HSSE	Health Safety Security and Environment
LCP	Local Control Panel
LCV	Level Control Valve
LP	Low Pressure
MSDS	Material Safety Data Sheet
MTPA	Metric Tonnes per Annum
MTPD	Metric Tonnes Per Day
NAG	Non-Associated Gas
NUPRC	Nigerian Upstream Petroleum Regulatory Commission
OML	Operating Mining License
OSD	Operational Shut Down
PAS	Process Automation System
PC	Pressure Controller
PCV	Pressure Control Valve
PTW	Permit to Work
SPDC	Shell Petroleum Development Company of Nigeria

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

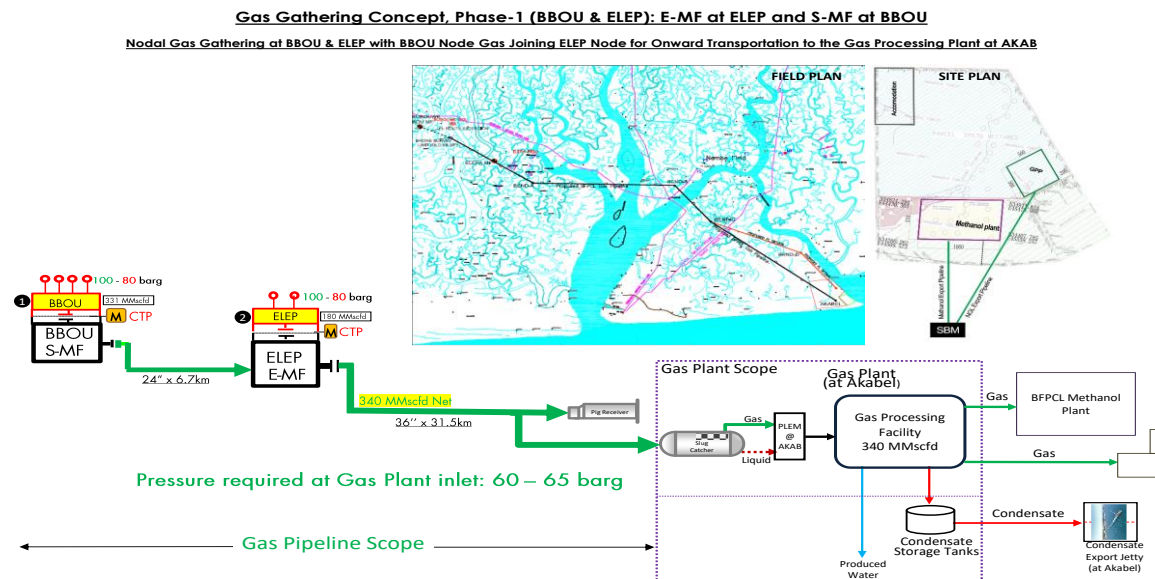


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

The purpose of this Performance Guarantee including Performance Test Procedure is to define the requirements for measuring performance on this project and specify the steps for carrying out performance Tests.

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

Document Number	Document Title
API 1104	Welding of Pipeline and Related Facilities
ASME B31.3	Process Piping
ASME B31.8	Gas Transmission and Distribution Piping Systems
AWS D1.1	Structural Welding
ASNT SNT-TC-1A	American Society for Non-Destructive Testing, recommended practice
ASTM A694	Carbon and Alloy Steel Forgings for Pipe Flanges. Fittings, Valves and Parts for High Pressure Transmission Service.
ASME B 16.34	Valves – Flanged Threaded and Welded Ends
API 598	valve Inspection and test

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Materials / Equipment Warranties

All OEMs and equipment vendors shall provide equipment warranty for a period to guarantee their equipment performance. For equipment that will be stored for some time before commissioning, the OEM shall provide preservation requirements to ensure the equipments are in good condition during the time of operation.

The warranties for Materials / equipment shall be effective for twelve (12) months after the commissioning or eighteen (18) months after delivery whichever occurs first.

However, for equipment packages like the Corrosion Inhibitor package, Supplier / OEM shall provide a 24-month warranty commencing from the date of commissioning.

If these cannot be obtained, CONTRACTOR will discuss internally and obtain specific instructions.

Purchase Orders will clearly indicate the Warranty requirements, and bid evaluations shall show vendor quoted warranty conditions which are provided in the quoted price.

4. Materials/Equipment Acceptance Tests

4.1. Factory Acceptance Test (FAT)

There shall be performance testing of major equipment in the OEM/vendor's factory, and it should be witnessed by the CONTRACTOR, COMPANY and Regulatory body (NUPRC) or their nominated 3rd party inspection agents.

The major equipment are Corrosion Inhibitor package, Metering Package, Actuated Valves, Control and Safeguarding Systems, Electrical Power systems, and Communication systems.

The extent of witness and resource allocation should vary depending on criticality/complexity of equipment and resource availability. This will be determined as part of approval of the test procedures.

Criticality Rating System and Assignment of Criticality Rating will be reviewed by BFPCL for any upgradation of Criticality Rating.

The FAT proposal shall be approved by COMPANY.

The OEM / Vendor will provide the technicians, test facilities and any special testing gear.

COMPANY representatives witnessing the factory acceptance tests will be required to indicate their acceptance/otherwise of the equipment on completion of the test. Where the equipment was not accepted, they shall document deficiencies noticed which necessitated its rejection so that the vendor can address them before shipment

For pre-commissioning in the fabrication yards, Vendor and CONTRACTOR will co-ordinate the preparation and execution of the test plans, and COMPANY shall approve.

4.2. Site Acceptance Test (SAT)

Site Acceptance Tests, which are essentially a repeat of the FAT's but done at site after installation, will be carried out during pre-commissioning and commissioning to confirm that the equipment still functions as specified after transportation and installation.

During SAT activities of the GGPL, the following guidelines and/or documentation will be prepared and/or followed.

- All data incorporated for maintenance routines, Operations HSE case, procedures, operating / maintenance manuals, and routines necessary to support operations shall be in place.
- A commissioning hand-over checklist (to be used in the hand-over process from CONTRACTOR to COMPANY) detailing outstanding items that may exist at hand-over phase from commissioning. They shall be identified as punch list items classified into categories 'A' and 'B' as follows:
- Category 'A' punch list items (Interim Handover) are items that must be repaired, completed

or corrected before the system can be handed-over or accepted.

- Category 'B' punch list items (Final Handover) will be items that can be repaired, completed, or corrected at a later date and allow the system to be operated without jeopardizing safety of people, asset, or the environment.
- The initial operating spares and special maintenance tools shall be available on site prior to hand-over. An audit shall be conducted on the status of all spare parts prior to handover.

5. Performance Tests

After the facilities have reached 'steady state' condition during commissioning, a 72-Hour Performance Test shall be carried out. The purpose of the 72-hour performance test is to confirm that the capability of the individual systems combined as a whole, fulfill the performance requirements specified in the functional specification, in all modes of operation.

Apart from Gas Pipelines, facilities would include GPP and Methanol Units, therefore Performance Tests shall be conducted when these facilities also have attained "Steady State".

All necessary process parameters shall be logged and recorded to enable the performance of each system to be documented for acceptance by COMPANY.

The run shall be uninterrupted and product gas quantity and quality shall be maintained continuously over the whole period. Any deviation from the performance requirements shall be rectified by CONTRACTOR and a further 72- hour performance test carried out.

To avoid unnecessary delays, the CONTRACTOR and COMPANY shall agree exact format of Performance Test within the Start-up and Operating constraints of the GGPL.

This document defines the procedure for undertaking the 72-Hour Performance Test of the plant. 72-Hour Performance Test of the GGPL shall be undertaken after the facilities have achieved steady state operation at design conditions, or such conditions as are achievable within the limitations of gas supply from SPDC (Seller) or gas export to GPP. Timing of such test shall be notified by the CONTRACTOR to COMPANY at least 48 hours before commencement of the test.

When CONTRACTOR is satisfied that the GGPL is ready for the 72-Hour Performance Test, CONTRACTOR will notify COMPANY of his intension to start the 72-Hour Performance Test. The test will commence 48 hours after such notice is given by the CONTRACTOR and accepted by Company.

5.1. Company Obligations and Responsibilities

During all tests, runs and re-runs, COMPANY responsibilities with respect to 72-hour Performance Test include the following.

General Obligations

COMPANY shall;

- assign a Representative(s) to witness the recording of operations and functional parameters,
- sign operating permits as necessary,
- advice when inlet gas can be admitted to the facility and the target flows, temperatures and pressures,

- approve the circuits that will not be in automatic mode, if any,
- sign the logbook and data collection sheets during the test.
- provide single point for operating instructions and approvals.

COMPANY is to review the 72-hour performance test report prepared by CONTRACTOR and either issue the Certificate of Initial Acceptance or advise CONTRACTOR in writing in event of any failure in the integrity of the performance test.

Feed Conditions

The operating conditions will be determined by the availability of production from SPDC. COMPANY to ensure that sufficient flow at the appropriate temperature and pressure is made available for the facility to run at design conditions.

Gas Export

The GPP must be available to run the GGPL at the design capacity. If the GPP is not available, the Pipelines cannot be commissioned because there is no recycle facility to allow continuous production of gas and continuous gas flaring is not permitted.

Operation

Pigging: No pigging will be done during the period of the test so that test can be carried out at stable operating conditions.

COMPANY shall not adjust the operating parameters during the test run so that test can be carried out at stable operating conditions.

5.2. CONTRACTOR Responsibilities

The CONTRACTOR is responsible for organisation and management of the 72 Hours Test and integration and supervision with COMPANY personnel.

The CONTRACTOR shall carry out the performance test and shall be responsible for the following:

- Maintain all necessary records for the test
- Log equipment failure and any other significant event happening during the test
- Rectify any problem during the test
- CONTRACTOR shall maintain a logbook of GGPL operations. The logbook shall record all GGPL operations and parameters and the time at which they occurred.

In the absence of COMPANY Representatives, the Contractor shall take full responsibility to gather the readings and present them to the COMPANY representative(s) at the earliest opportunity for his signature.

Times shall be recorded to the nearest minutes. Detailed notes of any differences in opinion,

between COMPANY and CONTRACTOR, of faults or causes of shutdowns shall be recorded separately and attached to the logbook.

At the end of the test, both COMPANY and CONTRACTOR Representatives shall sign and date the logbook. In the event of any dispute regarding the content of the logs, the CONTRACTOR and / or COMPANY opinions shall be noted in the logbook.

5.3. Successful Completion of 72-Hour Performance Test

72-Hour Performance Test shall be considered successful when the GGPL runs continuously for 72 hours, without any TECHNICAL INTERRUPTION, at the design capacity or at a constrained capacity due to external factors. The constrained capacity may be imposed during the test due to non-availability of gas flow at full capacity.

TECHNICAL INTERRUPTION is defined as shutdown of the facility during the test due to any technical fault attributable to the operation the facility. Failure of spared equipment where the facility operation can be continued with starting of the spare equipment shall not be considered as technical interruption. The facility shutdown caused by external factors not attributable to the operation of the facility shall not be considered as technical interruption.

5.4. Test Procedure

This section describes the 72-Hour Performance Test procedure. The test shall be conducted in accordance with the conditions and procedures specified below.

Note: If the GPP will not be available to export gas, the GGPL cannot be tested.

To monitor and record the activities during the 72 Hours Test, personnel will be stationed permanently at GGPL site. CONTRACTOR will arrange staff working hours accordingly and COMPANY Representatives will be invited to witness all tests.

During the test period, CONTRACTOR will operate the GGPL in accordance with the conditions stipulated by COMPANY.

A test report will be prepared by CONTRACTOR and issued to COMPANY within 2 weeks of completion of the 72-Hour Performance Test.

5.5. Parameters to Be Recorded and Evaluated

The parameters to be recorded will be compiled on reading sheets and the readings will be taken at four hourly intervals from the start of the 72-hour test.

The readings will be taken by the CONTRACTOR and witnessed by a designated COMPANY Representative. These shall include all mechanical, electrical and instrument readings required for the 72-hour test.

During the 72-Hour Performance Test, emphasis will be to run it continuously without technical interruption. The performance parameters are tested separately during the test to verify that the GGPL is meeting or capable of meeting the Performance Parameters.

Flowrate

The flowrate at the inlet of GGPL (Delivery Point) and the outlet (GPP) shall be measured to determine the deviation from the design conditions.

Other parameters

All instrument data going from GGPL to the central control room at GPMC are monitored and stored in the Control System. These data can be retrieved and trended over a period of time. Hence the parameters, which may be routinely monitored during the run, may be retrieved from the Control System.

During the run, the GGPL operating parameters like flowrate, pressures and temperatures are measured and compared with the design values to ensure the facility is running as per the design parameters.

A list of instruments, which may be monitored, for such test run will be prepared. The list will be an indication of major parameters but any instrument signal going to the central control room can be monitored and retrieved from the Control System.

Evaluation

Results will be evaluated on the following basis.

All measurements shall be averaged over the period of the test run.

If any measurements are demonstrably inconsistent with the bulk of the data, then data will be adjusted to achieve consistency and correctness in accordance with generally accepted engineering principles as mutually agreed between COMPANY and CONTRACTOR.

In the event that the conditions of feed or externally supplied utilities or ambient air differ from those on which the design is based, then the results of the test will be adjusted in accordance with the generally accepted engineering principle as mutually agreed between COMPANY and CONTRACTOR.

5.6. Constraints / Continuity

Any technical interruption of the GGPL due to reasons attributable to the CONTRACTOR will cause the test to be repeated.

Interruption of the test, which is attributed to COMPANY or with COMPANY agreement, shall not constitute failure of the test. Any hours clocked prior to such outage will therefore be accrued as

part of the 72-hour period.

If the reasons for interruption cannot be attributed to the CONTRACTOR, COMPANY and CONTRACTOR will discuss the cause and timing of the interruption and decide whether the run should be considered to be completed, or the run can continue after interruption or the run needs to be repeated.

Mechanical Equipment Failure

Failure of GGPL equipment, which follows with the termination of the 72-Hour Performance Test or deemed to jeopardize safety, will result in the test being repeated. Failure of spared equipment where the GGPL operation can be continued with starting of the spare equipment shall not be considered as technical interruption. In the event of failure of spared equipment, designated stand-by equipment will be used to maintain continuous running of the facility during the test.

Process Failure

Failure of the GGPL to meet the specifications of operation on a sustained basis may require the run to be repeated. The specifications are monitored by instruments in the Central Control System. In the event of such failure, COMPANY and CONTRACTOR will discuss the reasons for such failure and the corrective actions need to be taken. They will also decide whether the run can be continued or repeated after corrective action taken.

5.7. Acceptance Requirements

Successful completion of the 72-Hour Performance Test will be based on the following:

The GGPL must run continuously for 72 hours with the automatic shutdown devices in operations.

All instrumentation will be in automatic mode and overrides and /or by-passes will not be employed.

There will be no constant operator intervention to achieve normal operation of systems.

The CONTRACTOR and the designated COMPANY Representative will sign all reading sheets.

The CONTRACTOR will file these reading sheets for future reference.

The CONTRACTOR will issue a close out report with the copies of all reading sheets to COMPANY.

6. Reliability Testing

As a final test for HAND OVER of the facility from CONTRACTOR to COMPANY, the CONTRACTOR shall, immediately after completion of the 72-hour Performance test and issuance of the Certificate of Initial Acceptance by COMPANY, demonstrate the facilities reliability for a period of 3 months. The purpose of this is to confirm the reliability of the individual equipment items and of various facilities as a whole.

During the reliability test period, equipment failures shall be logged, measures to rectify the fault recorded and the total down-time logged.

The test procedure will follow a similar pattern to that of the Performance Test.

Equipment items to be monitored are:

Type A: Rotating equipment (corrosion inhibitor pumps). Trips caused by associated process controls and protection equipment are to be included in equipment down time.

Type B: Metering systems; process control systems.

Type C: ESD/OSD systems; fire & gas detection systems; UPS system.

For spared equipment, any individual item shall be placed on 'standby' for only one week at a time. This shall ensure that each item is in operation and tested for a significant part of the 3-month period. The reliability of each item shall be recorded separately.

The GGPL reliability test shall be accepted provided that the following availability percentages are achieved during the specified periods.

	Type A Equipment	Type B Equipment	Type C Equipment
Weeks 1-4	70%	80%	95%
Weeks 5-8	85%	90%	99%
Weeks 9-13	90%	95%	99.85%

6.1. Documentation

For the entire duration of the GGPL reliability test, COMPANY and the Contractor shall each assign full-time senior staff to the plant control room whose duty shall be to maintain a logbook of plant operations. The logbook shall record all facility operations and the time at which they occurred. Additionally, a separate detailed log shall be kept for each and every fault other than intentional shutdowns required by COMPANY's operations programme. Times shall be recorded to the nearest minute. Detailed notes of any differences in opinion between COMPANY and the contractor of the causes of faults or shutdowns shall be recorded separately and attached to the

logbook.

At the end of each shift, the COMPANY and the Contractor representatives shall both sign and date the logbook before handing over to the next shift. In the event of any dispute regarding the content of the logs, the contractor's opinion shall also be noted in the logbooks.

6.2. Hand-Over to COMPANY

At the end of a successful reliability test, COMPANY shall issue the final Acceptance Certificate to CONTRACTOR relieving CONTRACTOR of the responsibility of running the facilities and handing the responsibility over to COMPANY.